

might suggest. But it can make stock market losses worse than what the past shows.

Wars are fractal in nature. A war that kills more people than the devastating Second World War is possible—not likely, but not a zero probability, although such a war has never happened in the past.

Second, I will introduce an illustration from nature that will help to make the point about precision. A mountain is somewhat similar to a stone: it has an affinity with a stone, a family resemblance, but it is not identical. The word to describe such resemblances is *self-affine*, not the precise *self-similar*, but Mandelbrot had trouble communicating the notion of affinity, and the term *self-similar* spread with its connotation of precise resemblance rather than family resemblance. As with the mountain and the stone, the distribution of wealth above \$1 billion is not exactly the same as that below \$1 billion, but the two distributions have “affinity.”

Third, I said earlier that there have been plenty of papers in the world of econophysics (the application of statistical physics to social and economic phenomena) aiming at such calibration, at pulling numbers from the world of phenomena. Many try to be predictive. Alas, we are not able to predict “transitions” into crises or contagions. My friend Didier Sornette attempts to build predictive models, which I love, except that I can not use them to make predictions—but please don’t tell him; he might stop building them. That I can’t use them as he intends does not invalidate his work, it just makes the interpretations require broad-minded thinking, unlike models in conventional economics that are fundamentally flawed. We may be able to do well with some of Sornette’s phenomena, but not all.

WHERE IS THE GRAY SWAN?

I have written this entire book about the Black Swan. This is not because I am in love with the Black Swan; as a humanist, I hate it. I hate most of the unfairness and damage it causes. Thus I would like to eliminate many Black Swans, or at least to mitigate their effects and be protected from them. Fractal randomness is a way to reduce these surprises, to make some of the swans appear possible, so to speak, to make us aware of their consequences, to make them gray. *But fractal randomness does not yield precise answers.* The benefits are as follows. If you know that the stock market *can* crash, as it did in 1987, then such an event is not a Black Swan. The crash of 1987 is not an outlier if you use a fractal with an exponent of three. If you know that biotech companies can deliver a megablockbuster drug, bigger than all we’ve had so far, then it won’t be a Black Swan, and you will not be surprised, should that drug appear.